

Signal Processing

Exercise Sheet 5 - Solution

SoSe26

Question 1 – Sampling Frequency and Aliasing

The continuous-time signals $x_1(t)$ and $x_2(t)$ are given by

$$x_1(t) = \sin(2\pi 100t) \quad (3)$$

and

$$x_2(t) = \cos(4000\pi t), \quad (4)$$

respectively. For case of simplicity, we assume both signals to be T_s -spaced sampled once again.

In practice, the effect aliasing needs to be addressed. It is for this reason, that signals are often sampled at rates much higher than twice the highest frequency component contained in the signal of interest. Determine, if aliasing might turn out to be in issue in case the sampling time for $x_1(t)$ and $x_2(t)$ is chosen to be

$$T_1 = \frac{1}{400} \text{ s}$$

and

$$T_2 = \frac{1}{1500} \text{ s},$$

respectively.

Optional: Verify your results by sketching the frequency spectra of the discrete-time signals $x_1(n)$ and $x_2(n)$.

Solution:

Since we know that $x_n = \sin(2\pi f t)$, we can directly determine the frequencies of $x_1(t)$ and $x_2(t)$ to be

$$f_{x1} = 100 \text{ Hz}$$

and

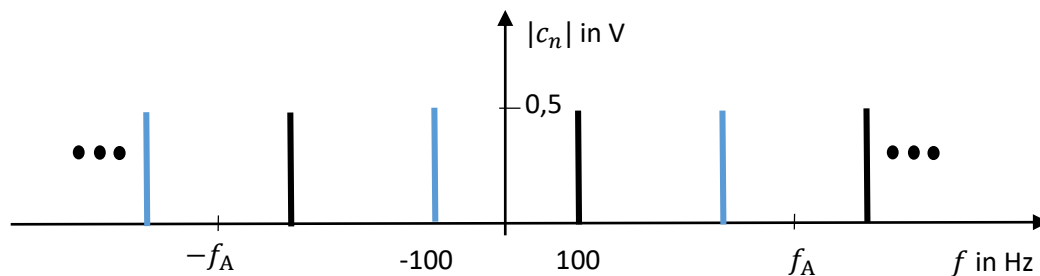
$$f_{x2} = 2000 \text{ Hz},$$

respectively.

In our case, the sampling frequency $f_1 = \frac{1}{T_1} = 400 \text{ Hz}$ holds for $f_1 > 2 \cdot f_{x1}$. It is for this reason, that aliasing will not be an issue for $x_1(t)$.

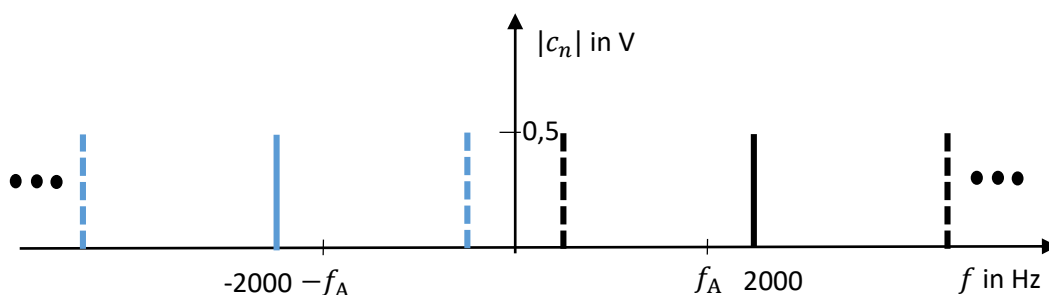
However, by choosing $f_2 = \frac{1}{T_2} = 1500 \text{ Hz}$, aliasing will turn out to be an issue for $x_2(t)$ since $f_2 < 2 \cdot f_{x2}$.

$x_1(t) = \sin(2 \pi \cdot 100 \cdot t)$: No DC-Component. This means $c_0 = 0$.



No aliasing since one cannot find any additional components in the baseband (-100 Hz to 100 Hz).

$x_2(t) = \cos(4000 \pi t)$. No DC-offset. This means $c_0 = 0$.



Aliasing since additional components (dashed lines) appear in the baseband (-2000 Hz to 2000 Hz).

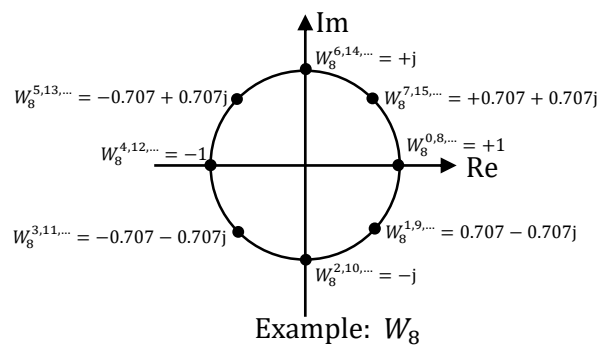
Part 2 – Discrete Fourier Transform (DFT)

The DFT allows for the numerical calculation of the spectrum of a signal. Keep in mind that the amount of acquired signal values N corresponds to the same amount of calculated frequency components M .

Discrete Fourier Transform

$$X_{\text{DFT}}[m] = \sum_{n=0}^{N-1} x[n] e^{-jn2\pi m/N} = \sum_{n=0}^{N-1} x[n] \cdot W_N^{nm},$$

where $W_N = e^{-\frac{j2\pi}{N}}$ is the rotation phasor
and $m = 0, \dots, N - 1$.



Inverse Discrete Fourier Transform

$$x[n] = \frac{1}{N} \sum_{m=0}^{N-1} X_{\text{DFT}}[m] \cdot W_N^{-nm},$$

where $m = 0, \dots, N - 1$.

Question 1

The sequences $x_0[n]$, $x_1[n]$ and $x_2[n]$ are given by

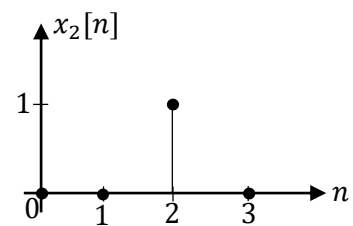
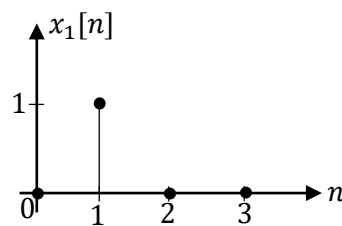
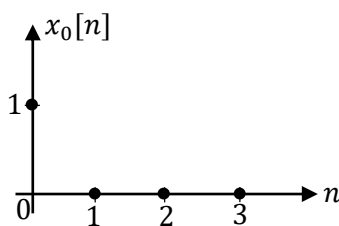
$$x_0[n] = \delta[n],$$

$$x_1[n] = \delta[n - 1],$$

$$x_2[n] = \delta[n - 2],$$

where $n = 0, \dots, N - 1$.

a) Sketch the sequences for $N = 4$.

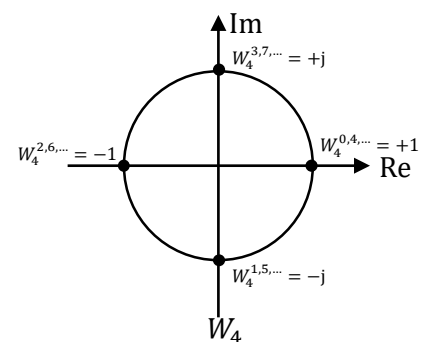


b) Calculate the DFT_4 for each sequence.

$$X_0[m] = \sum_{n=0}^3 \delta[n] e^{-jn2\pi m/4} = [1, 1, 1, 1]$$

$$X_1[m] = \sum_{n=0}^3 \delta[n - 1] e^{-jn2\pi m/4} = [1, -j, -1, j]$$

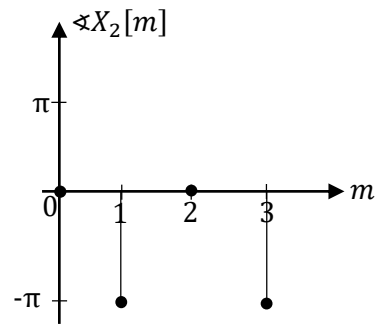
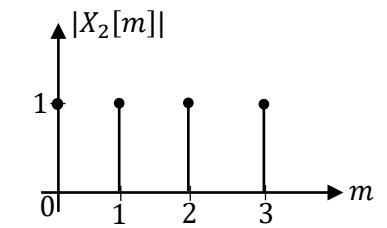
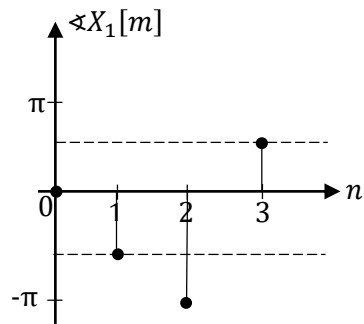
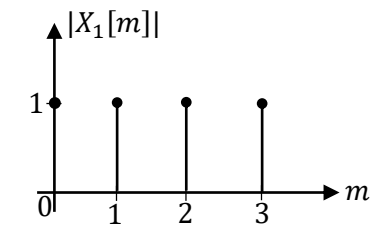
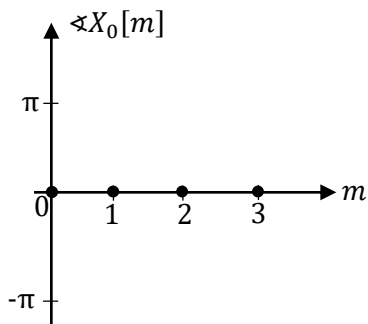
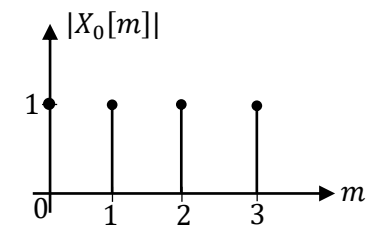
$$X_2[m] = \sum_{n=0}^3 \delta[n - 2] e^{-jn2\pi m/4} = [1, -1, 1, -1]$$



Hint: In depth solution for $X_1[1]$

$$X_1[1] = \delta[-1]W_4^{0 \cdot 1} + \delta[0]W_4^{1 \cdot 1} + \delta[1]W_4^{2 \cdot 1} + \delta[2]W_4^{3 \cdot 1} = 0 \cdot 1 + 1 \cdot (-j) + 0 \cdot (-1) + 0 \cdot (j) = -j$$

c) Sketch the magnitude and phase of $X_0[m]$, $X_1[m]$ and $X_2[m]$, respectively.



Question 2

The impulse response $h[n]$ and the input signal $x[n]$ of an LTI system are given by

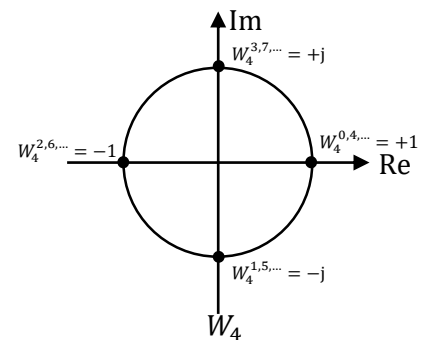
$$h[n] = [2, 0, -1, 1]$$

and

$$x[n] = [3, 4, 2, 1].$$

a) Determine the DFT_4 of $h[n]$ and $x[n]$.

$$\begin{aligned} H[m] &= \sum_{n=0}^{N-1} W_N^{nm} \cdot h[n] = \sum_{n=0}^3 W_4^{nm} \cdot h[n] \\ H[m] &= \begin{pmatrix} W_4^{0 \cdot 0} & W_4^{1 \cdot 0} & W_4^{2 \cdot 0} & W_4^{3 \cdot 0} \\ W_4^{0 \cdot 1} & W_4^{1 \cdot 1} & W_4^{2 \cdot 1} & W_4^{3 \cdot 1} \\ W_4^{0 \cdot 2} & W_4^{1 \cdot 2} & W_4^{2 \cdot 2} & W_4^{3 \cdot 2} \\ W_4^{0 \cdot 3} & W_4^{1 \cdot 3} & W_4^{2 \cdot 3} & W_4^{3 \cdot 3} \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 0 \\ -1 \\ 1 \end{pmatrix} = \\ &= \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3+j \\ 0 \\ 3-j \end{pmatrix} \end{aligned}$$



$$\begin{aligned} X[m] &= \sum_{n=0}^{N-1} W_N^{nm} \cdot x[n] = \sum_{n=0}^3 W_4^{nm} \cdot x[n] \\ X[m] &= \begin{pmatrix} W_4^{0 \cdot 0} & W_4^{1 \cdot 0} & W_4^{2 \cdot 0} & W_4^{3 \cdot 0} \\ W_4^{0 \cdot 1} & W_4^{1 \cdot 1} & W_4^{2 \cdot 1} & W_4^{3 \cdot 1} \\ W_4^{0 \cdot 2} & W_4^{1 \cdot 2} & W_4^{2 \cdot 2} & W_4^{3 \cdot 2} \\ W_4^{0 \cdot 3} & W_4^{1 \cdot 3} & W_4^{2 \cdot 3} & W_4^{3 \cdot 3} \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \\ 2 \\ 1 \end{pmatrix} = \\ &= \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 \\ 1-3j \\ 0 \\ 1+3j \end{pmatrix} \end{aligned}$$

b) Determine the DFT $Y[m]$ of the output signal $y[n]$ of the system.

$$Y[m] = X[m] \circ H[m] = \begin{pmatrix} 10 \\ 1 - 3j \\ 0 \\ 1 + 3j \end{pmatrix} \circ \begin{pmatrix} 2 \\ 3 + j \\ 0 \\ 3 - j \end{pmatrix} = \begin{pmatrix} 20 \\ 6 - 8j \\ 0 \\ 6 + 8j \end{pmatrix}$$